Portfolio Optimization

Financial Data: I.I.D. Modeling

Daniel P. Palomar (2025). *Portfolio Optimization: Theory and Application.*
Cambridge University Press.

portfoliooptimizationbook.com

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Outline

1. I.I.D. Model
2. Sample Estimators
3. Location Estimators
4. Gaussian ML Estimators
5. Heavy-Tailed ML Estimators
6. Prior Information
   - Shrinkage
   - Factor Models
   - Black-Litterman Model
7. Summary

Executive Summary

- The efficient-market hypothesis suggests that a security's price reflects its intrinsic value, incorporating all available information.
- Then, prices can be modeled as a random walk with returns being independent and identically distributed (i.i.d.) random variables.
- These slides explore various methods to characterize the multivariate i.i.d. distribution of returns.
- Methods range from simple sample estimators to more advanced robust non-Gaussian estimators.
- Advanced estimators incorporate prior information through:
  - Shrinkage techniques
  - Factor modeling
  - Prior views
- Reference: (Palomar 2025, chap. 3)

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I.I.D. Model

Introduction to Financial Data Modeling

- Financial data modeling is crucial for understanding and predicting market behaviors.
- It involves N securities or assets from various classes (e.g., bonds, equities, commodities).

Random Returns Representation

- Random returns of assets at time t are denoted by $\boldsymbol{x}_t$ or $\boldsymbol{r}_t$.
- The time index t can represent different periods (minutes to years).

I.I.D. Model for Returns

- Returns are modeled as:

$$\boldsymbol{x}_t = \boldsymbol{\mu} + \boldsymbol{\epsilon}_t.$$

- $\boldsymbol{\mu}$ represents the expected return, and $\boldsymbol{\epsilon}_t$ is the residual with zero mean.
- $\boldsymbol{\Sigma}$ denotes the covariance matrix of residuals.

I.I.D. Model

Efficient-Market Hypothesis

- The i.i.d. model is motivated by the efficient-market hypothesis (EMH).
- Eugene F. Fama, a proponent of EMH, won the Nobel Prize in 2013.

Random Walk Model

- The i.i.d. model corresponds to the random walk model on log-prices $\boldsymbol{y}_t$.
- Log-prices are defined as: $\boldsymbol{y}_t \triangleq \log \boldsymbol{p}_t$.
- This leads to the i.i.d. model when considering log-returns: $\boldsymbol{x}_t = \boldsymbol{y}_t - \boldsymbol{y}_{t-1}$.

Limitations of the I.I.D. Model

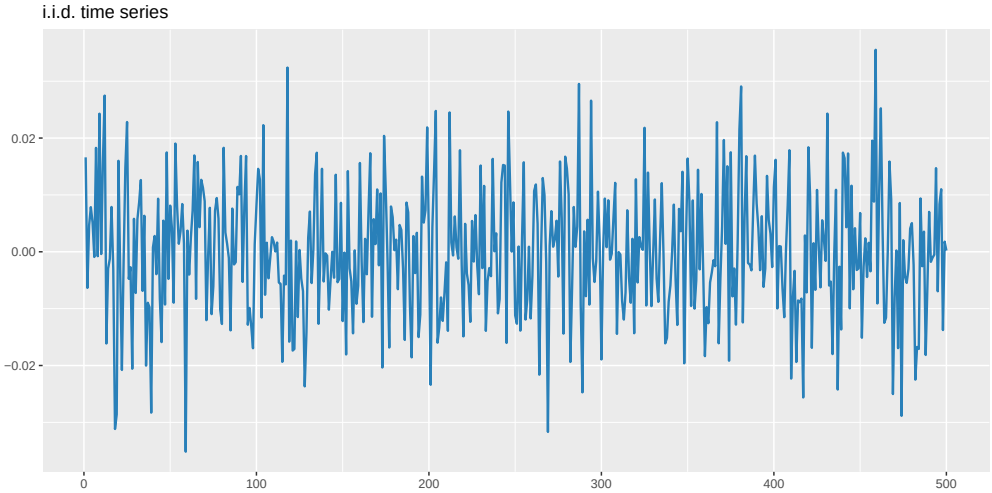
- It ignores temporal structure or dependency in financial data.

Sophisticated Time Series Models

- There is a myriad of different time series models developed over the past seven decades that attempt to capture the temporal structure.
- Recommended textbooks for financial data modeling are: (Meucci 2005), (Tsay 2010), (Ruppert and Matteson 2015), (Lütkepohl 2007), (Tsay 2013).

I.I.D. model

Example of a synthetic Gaussian i.i.d. time series:



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Sample Estimators

Estimating I.I.D. Model Parameters

- The parameters $(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ are estimated using historical data $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_T$.
- It utilizes T past observations for estimation.

Sample Mean Estimator:

$$\hat{\boldsymbol{\mu}} = \frac{1}{T} \sum_{t=1}^{T} \boldsymbol{x}_t.$$

- It represents the average of past observations.

Sample Covariance Matrix Estimator:

$$\hat{\boldsymbol{\Sigma}} = \frac{1}{T-1} \sum_{t=1}^{T} (\boldsymbol{x}_t - \hat{\boldsymbol{\mu}})(\boldsymbol{x}_t - \hat{\boldsymbol{\mu}})^{\mathsf{T}}.$$

- It measures variability around the sample mean.

Unbiasedness of Estimators

- The sample mean and covariance estimators are unbiased.
- The expected values of $\hat{\boldsymbol{\mu}}$ and $\hat{\boldsymbol{\Sigma}}$ equal the true values $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$.

Sample Estimators

Bias in Sample Covariance With $1/T$

- Using $1/T$ instead of $1/(T-1)$ in covariance estimation introduces bias.
- It results in an underestimate: $\mathbb{E}[\hat{\mathbf{\Sigma}}] = \left(1 - \frac{1}{T}\right)\mathbf{\Sigma}$.

Consistency of Estimators

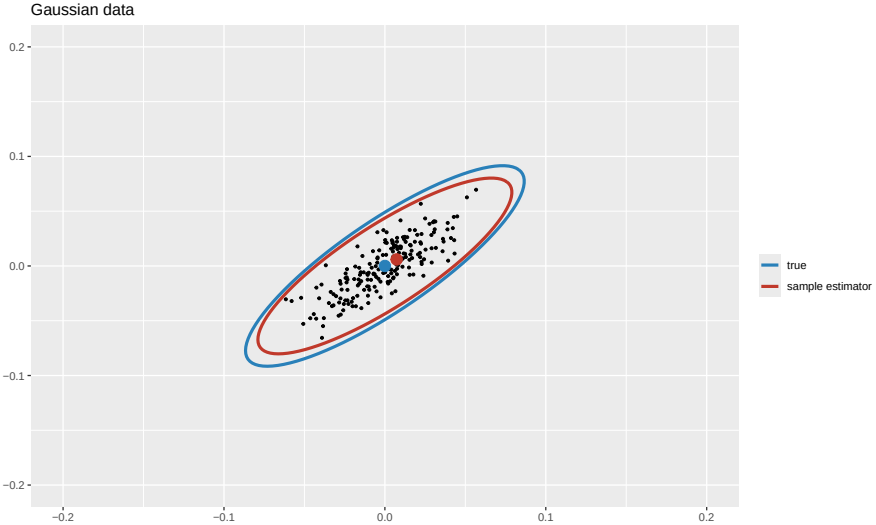
- Both estimators are consistent, converging to true values as $T \to \infty$.
- This is supported by the law of large numbers.

Estimation Error Reduction

- The estimation error decreases with increasing T.
- This is illustrated next through synthetic Gaussian data with $N = 100$.
- The normalized error approaches zero as the sample size grows.

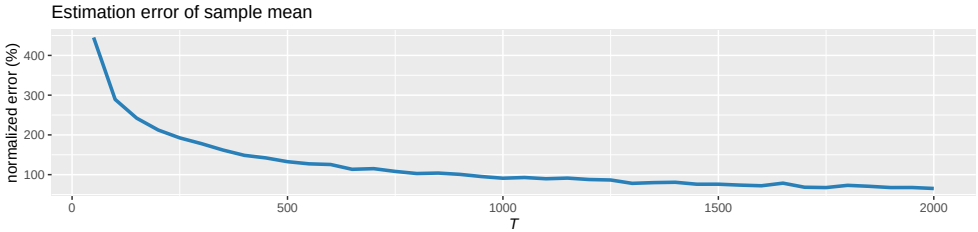
Sample Estimators

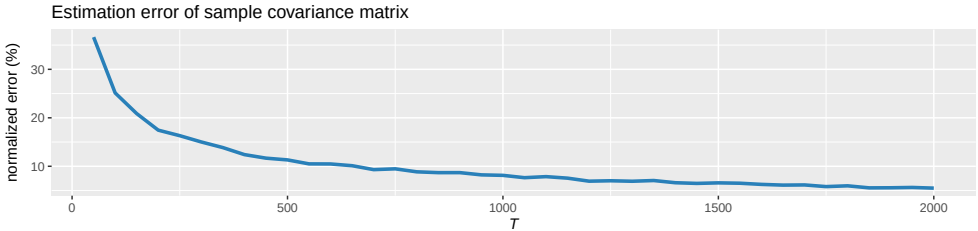
Illustration of sample mean and sample covariance matrix:



Sample Estimators

Estimation error of sample estimators versus number of observations (for Gaussian data with $N = 100$):





Sample Estimators

Challenges With Sample Estimators

- Simple and cost-effective but require a large number of observations T for accuracy.
- The sample mean $\hat{\boldsymbol{\mu}}$ is particularly inefficient, leading to noisy estimates.

High Estimation Error With Limited Data

- For $N = 100$ and $T = 500$, the normalized error of $\hat{\boldsymbol{\mu}}$ can exceed 100%.
- In words: the error magnitude is as large as the true value of $\boldsymbol{\mu}$.

Practical Limitations for Large T

- Lack of available historical data: Ideal data span (e.g., 20 years for $N = 500$) often exceeds available records.
- Lack of stationarity: Financial markets evolve, making long-term historical data less relevant.

Implications for Portfolio Optimization

- Limited data leads to noisy estimates of $\hat{\boldsymbol{\mu}}$ and $\hat{\boldsymbol{\Sigma}}$.
- Estimation noise undermines the reliability of portfolio designs.
- It challenges the practical adoption of Markowitz's portfolio theory.

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Location Estimators

Interpreting μ in the I.I.D. Model

- μ represents the central location of the distribution of random points.

Estimating the Central Location

- Various methods exist beyond classical sample estimators.
- Sample estimators are sensitive to extreme values and missing data.

Robust Multivariate Location Estimators

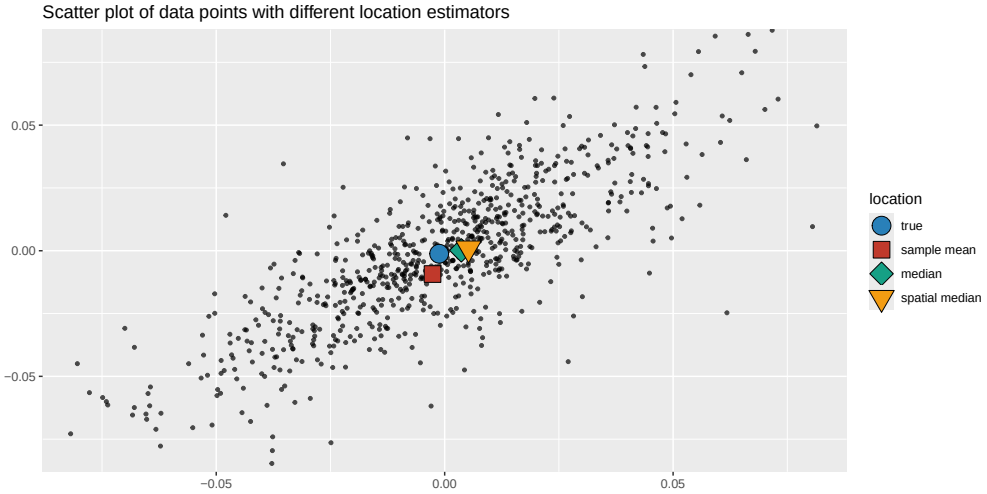
- Developed due to the limitations of sample estimators and least squares.
- Aim to be less affected by outliers and missing values.
- Historical research dates back to the 1960s.

Some Methods to Estimate This Center

- Classical approach: least squares (LS).
- Median estimator.
- Spatial median estimator.

Location Estimators

Illustration of different location estimators:



Least Squares (LS) Estimator

Least Squares (LS) Estimator

- It originates from Gauss's work in 1795 on planetary motions.
- It involves minimizing the squared difference between observed and predicted values.
- It is formulated as:

$$\underset{\boldsymbol{x}}{\text{minimize}} \quad \|\boldsymbol{y} - \boldsymbol{A}\boldsymbol{x}\|_2^2$$

- The closed-form solution is: $\boldsymbol{x}^\star = \left(\boldsymbol{A}^\mathsf{T}\boldsymbol{A}\right)^{-1}\boldsymbol{A}^\mathsf{T}\boldsymbol{y}$.

Application to the I.I.D. Model

- Estimating $\boldsymbol{\mu}$ in the i.i.d. model can be seen as an LS problem:

$$\underset{\boldsymbol{\mu}}{\text{minimize}} \quad \sum_{t=1}^{T} \|\boldsymbol{x}_t - \boldsymbol{\mu}\|_2^2$$

- The solution coincides with the sample mean:

$$\hat{\boldsymbol{\mu}} = \frac{1}{T}\sum_{t=1}^{T} \boldsymbol{x}_t.$$

Least Squares (LS) Estimator

Challenges With the Sample Mean

- It lacks robustness against outliers and heavy-tailed distributions.
- The vulnerability to contaminated points affects the reliability of $\hat{\boldsymbol{\mu}}$.

Robust Estimation Needs

- Due to the limitations of LS and the sample mean, alternative robust estimators have been explored.
- They aim to improve reliability and accuracy in the presence of outliers and non-normal distributions.
- The study of robust multivariate location estimators dates back to the 1960s.

Median Estimator

Median as a Robust Estimator

- The median separates the higher half from the lower half of a sample.
- It is considered the "middle" value, providing a typical representation of the data.
- It is unaffected by extreme values, unlike the mean.
- It provides robustness against outliers.
- It represents a more "typical" value of the dataset.

Multivariate Median

- There are multiple ways to extend the median to a multivariate setting.
- Elementwise median is a straightforward extension.

Elementwise Median in the I.I.D. Model

- It can be formulated as the optimization problem:

$$\underset{\boldsymbol{\mu}}{\text{minimize}} \quad \sum_{t=1}^{T} \|\boldsymbol{x}_t - \boldsymbol{\mu}\|_1$$

- It uses the ℓ_1-norm to measure error, offering robustness against outliers.

Spatial Median Estimator

Spatial or Geometric Median

- An extension of the univariate median to multivariate data.
- It is formulated as the optimization problem:

$$\underset{\boldsymbol{\mu}}{\text{minimize}} \quad \sum_{t=1}^{T} \|\boldsymbol{x}_t - \boldsymbol{\mu}\|_2$$

- It uses the ℓ_2-norm (Euclidean distance) as the measure of error.

Characteristics of the Spatial Median

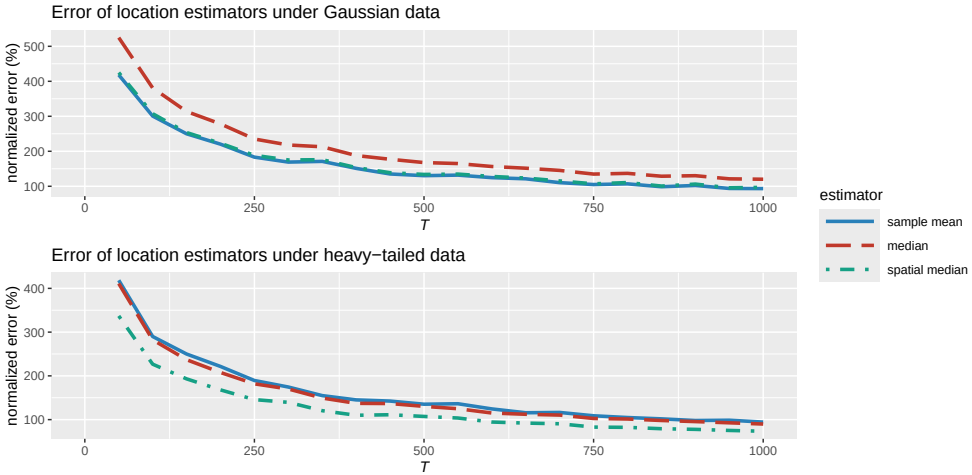
- The estimator for each element is not independent of other elements.
- For $N = 1$, it coincides with the univariate median.

Solving for the Spatial Median

- The problem is a convex second-order cone problem (SOCP).
- It can be solved using SOCP solvers or iterative algorithms.
- Efficient iterative algorithms use the majorization-minimization (MM) method (Sun, Babu, and Palomar 2017).

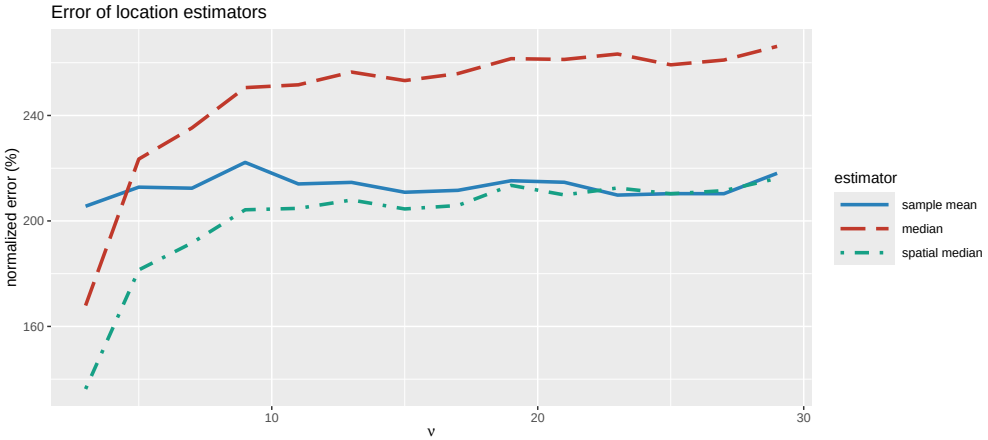
Numerical Experiments

Estimation error of location estimators versus number of observations (with $N = 100$):



Numerical Experiments

Estimation error of location estimators versus degrees of freedom in a t distribution (with $T = 200$ and $N = 100$):



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Preliminaries on ML Estimation

Maximum Likelihood Estimation (MLE)

- A fundamental technique in estimation theory.
- It involves selecting the parameter vector $\boldsymbol{\theta}$ that maximizes the likelihood of observing a given set of data.

Concept of MLE

- It is based on the probability distribution function (pdf) f of the random variable $\boldsymbol{x}$.
- For T independent observations $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_T$, the likelihood is $f(\boldsymbol{x}_1) \times \cdots \times f(\boldsymbol{x}_T)$.
- MLE chooses the parameter $\boldsymbol{\theta}$ that maximizes this product for a family of distributions $f_{\boldsymbol{\theta}}$.

Optimization Problem in MLE

- It is formulated as:

$$\underset{\boldsymbol{\theta}}{\text{maximize}} \quad f_{\boldsymbol{\theta}}(\boldsymbol{x}_1) \times \cdots \times f_{\boldsymbol{\theta}}(\boldsymbol{x}_T).$$

- Equivalently, maximizing the log-likelihood:

$$\underset{\boldsymbol{\theta}}{\text{maximize}} \quad \sum_{t=1}^{T} \log f_{\boldsymbol{\theta}}(\boldsymbol{x}_t).$$

Preliminaries on ML Estimation

Theoretical Properties of MLE

- Asymptotically unbiased: The MLE's bias diminishes as $T \to \infty$.
- Asymptotically efficient: It achieves the Cramer-Rao bound, representing the lowest variance for an unbiased estimator.

Practical Considerations

- The effectiveness of MLE's asymptotic properties depends on the size of T.
- Determining how large T needs to be for these properties to hold is crucial in practice.

Gaussian ML Estimation

PDF for I.I.D. Model with Gaussian Residuals

- Assuming residuals follow a multivariate normal distribution:

$$f(\boldsymbol{x}) = \frac{1}{\sqrt{(2\pi)^N |\boldsymbol{\Sigma}|}} \exp\left(-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^\mathsf{T}\boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})\right).$$

- The model parameters are $\boldsymbol{\theta} = (\boldsymbol{\mu}, \boldsymbol{\Sigma})$.

MLE Formulation for Gaussian I.I.D. Model

$$\underset{\boldsymbol{\mu},\boldsymbol{\Sigma}}{\text{minimize}} \quad \log\det(\boldsymbol{\Sigma}) + \frac{1}{T}\sum_{t=1}^{T}(\boldsymbol{x}_t-\boldsymbol{\mu})^\mathsf{T}\boldsymbol{\Sigma}^{-1}(\boldsymbol{x}_t-\boldsymbol{\mu}).$$

Gaussian ML Estimation

Deriving MLE for μ and Σ

- Set the gradient with respect to $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ to zero.
- This results in the estimators:

$$\hat{\boldsymbol{\mu}} = \frac{1}{T}\sum_{t=1}^{T}\boldsymbol{x}_t$$

$$\hat{\boldsymbol{\Sigma}} = \frac{1}{T}\sum_{t=1}^{T}(\boldsymbol{x}_t - \boldsymbol{\mu})(\boldsymbol{x}_t - \boldsymbol{\mu})^{\mathsf{T}}.$$

Comparison With Sample Estimators

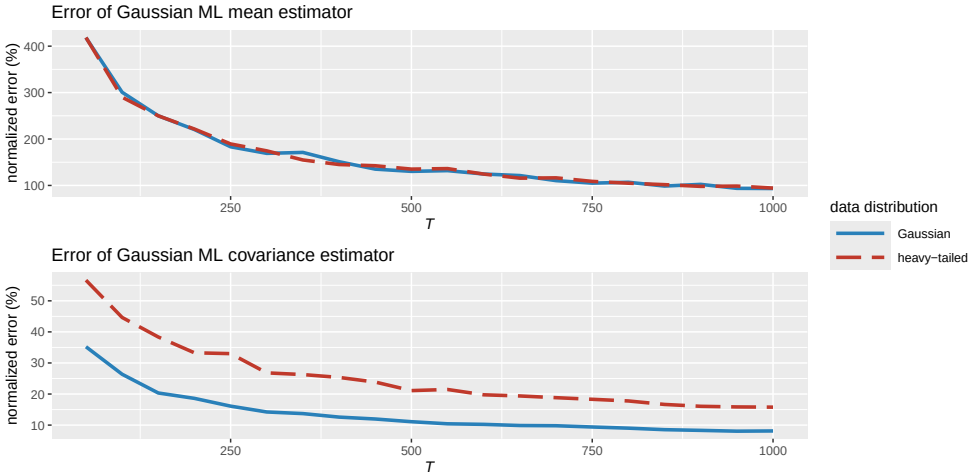
- The MLE coincides with the sample mean and covariance estimators, except for the factor $1/T$ instead of $1/(T-1)$.
- The MLE of the covariance matrix is biased but asymptotically unbiased as $T \to \infty$.

Implications of MLE

- Sample estimators are optimal for Gaussian-distributed data.
- For non-Gaussian distributions, the optimal ML estimators will be different.

Numerical Experiments

Estimation error of Gaussian ML estimators versus number of observations (with $N = 100$):



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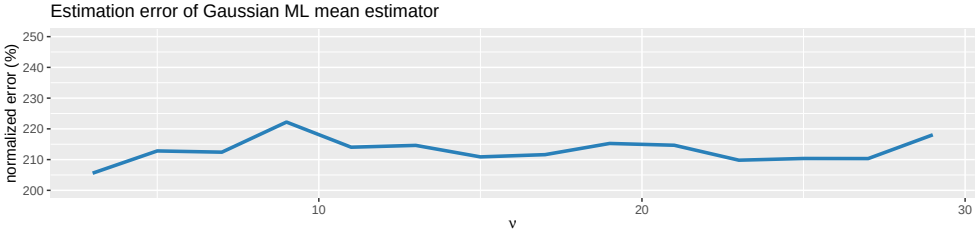

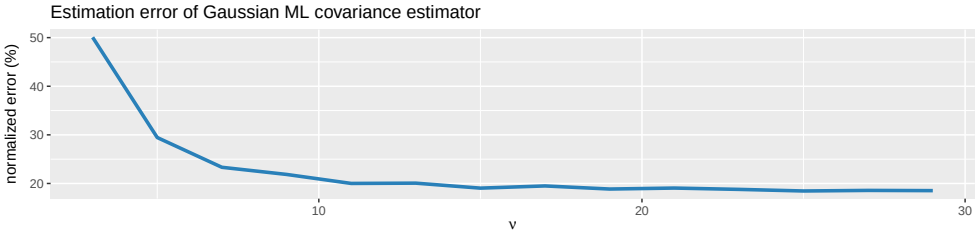

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Heavy-Tailed ML Estimators

Gaussian ML Estimators and Heavy-Tailed Distributions

- Optimal for Gaussian-distributed data, but financial data often exhibit heavy tails.
- There is a need to assess the impact of heavy-tailed distributions on these estimators.

Properties of Gaussian ML Estimators

- They coincide with sample estimators for Gaussian data.
- They are unbiased and consistent, which are beneficial characteristics.

Evaluating Estimator Performance With Heavy Tails

- Despite being unbiased and consistent, they may not be the best choice for heavy-tailed data.
- It is important to consider the efficiency and robustness of estimators under such conditions.
- There is potential for improved estimators that better handle the peculiarities of financial data distributions.

The Failure of Gaussian ML Estimators

Impact of Heavy Tails on Estimation

- Heavy tails significantly affect covariance matrix estimation but not mean estimation.
- The error varies with tail heaviness: smaller ν means heavier tails and larger estimation error.

Visualizing Detrimental Effects

- The following scatter plots illustrate the impact of heavy tails and outliers on estimation.
- With $T = 10$ and $N = 2$, the sample covariance matrix is sensitive to these effects.

Consequences for Gaussian ML Estimators

- They perform well for Gaussian data but poorly with outliers or heavy-tailed data.
- Due its simplificty, the sample covariance matrix is widely used by practitioners.

Challenges in Practice

- The prevalence of heavy-tailed distributions in financial data complicates estimation.
- A single outlier can lead to significantly skewed estimations of the covariance matrix.
- There is a need for more robust estimation techniques that can handle outliers and non-Gaussian distributions effectively.

The Failure of Gaussian ML Estimators

Effect of heavy tails and outliers in the Gaussian ML covariance matrix estimator:

Heavy-Tailed ML Estimation

MLE for Heavy-Tailed Distributions

- Gaussian MLE is not optimal for heavy-tailed data, which is common in finance.
- The Student t distribution is used to model heavy tails with parameter ν.

PDF for Multivariate t Distribution

- The probability density function is:

$$f(\boldsymbol{x}) = \frac{\Gamma((\nu+N)/2)}{\Gamma(\nu/2)\sqrt{(\nu\pi)^N|\boldsymbol{\Sigma}|}} \left(1 + \frac{1}{\nu}(\boldsymbol{x}-\boldsymbol{\mu})^\mathsf{T}\boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})\right)^{-(\nu+N)/2}$$

- The parameters are $\boldsymbol{\mu}$ (location), $\boldsymbol{\Sigma}$ (scatter matrix), and ν (degrees of freedom).

MLE Formulation With t Distribution

- For fixed $\nu = 4$, the MLE problem simplifies to:

$$\underset{\boldsymbol{\mu},\boldsymbol{\Sigma}}{\text{minimize}} \quad \log\det(\boldsymbol{\Sigma}) + \frac{\nu+N}{T}\sum_{t=1}^{T}\log\left(1 + \frac{1}{\nu}(\boldsymbol{x}_t-\boldsymbol{\mu})^\mathsf{T}\boldsymbol{\Sigma}^{-1}(\boldsymbol{x}_t-\boldsymbol{\mu})\right)$$

Heavy-Tailed ML Estimation

Deriving MLE for $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$

- Set the gradient with respect to $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ to zero.
- This results in the fixed-point equations for $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$:

$$\boldsymbol{\mu} = \frac{\frac{1}{T}\sum_{t=1}^{T} w_t(\boldsymbol{\mu}, \boldsymbol{\Sigma}) \times \boldsymbol{x}_t}{\frac{1}{T}\sum_{t=1}^{T} w_t(\boldsymbol{\mu}, \boldsymbol{\Sigma})}$$

$$\boldsymbol{\Sigma} = \frac{1}{T}\sum_{t=1}^{T} w_t(\boldsymbol{\mu}, \boldsymbol{\Sigma}) \times (\boldsymbol{x}_t - \boldsymbol{\mu})(\boldsymbol{x}_t - \boldsymbol{\mu})^{\mathsf{T}}$$

where the weights $w_t(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ are defined as

$$w_t(\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{\nu + N}{\nu + (\boldsymbol{x}_t - \boldsymbol{\mu})^{\mathsf{T}}\boldsymbol{\Sigma}^{-1}(\boldsymbol{x}_t - \boldsymbol{\mu})}.$$

Advantages of Heavy-Tailed MLE

- Provides a more robust estimation for datasets with heavy tails.
- Weights observations differently, reducing the influence of outliers.

Heavy-Tailed ML Estimation

MM-based method to solve the heavy-tailed ML fixed-point equations

Initialization:

- Choose initial point $(\boldsymbol{\mu}^0, \boldsymbol{\Sigma}^0)$.
- Set iteration counter $k \leftarrow 0$.

Repeat (kth iteration):

1. Iterate the weighted sample mean and sample covariance matrix as

$$\boldsymbol{\mu}^{k+1} = \frac{\frac{1}{T}\sum_{t=1}^{T} w_t(\boldsymbol{\mu}^k, \boldsymbol{\Sigma}^k) \times \boldsymbol{x}_t}{\frac{1}{T}\sum_{t=1}^{T} w_t(\boldsymbol{\mu}^k, \boldsymbol{\Sigma}^k)}$$

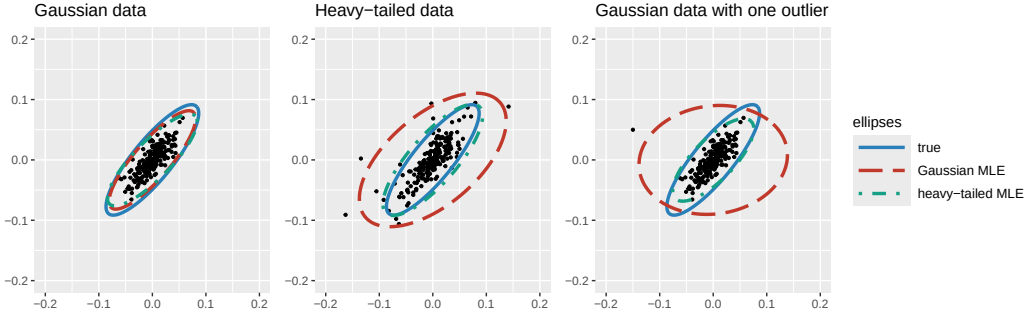
$$\boldsymbol{\Sigma}^{k+1} = \frac{1}{T}\sum_{t=1}^{T} w_t(\boldsymbol{\mu}^{k+1}, \boldsymbol{\Sigma}^k) \times (\boldsymbol{x}_t - \boldsymbol{\mu}^{k+1})(\boldsymbol{x}_t - \boldsymbol{\mu}^{k+1})^{\mathsf{T}}$$

2. $k \leftarrow k + 1$

Until: convergence

Numerical Experiments

Effect of heavy tails and outliers in heavy-tailed ML covariance matrix estimator:



Robust Estimators

Robust Estimators Overview

- Estimators resilient to outliers and deviations from the assumed distribution.
- They are reliable under non-ideal conditions.
- References: (Huber 1964), (Maronna 1976), (Maronna, Martin, and Yohai 2006), (Huber 2011), (Wiesel and Zhang 2014), (Zoubir et al. 2018, chap. 4), (Palomar 2025, chap. 3).

Measuring Robustness

- Influence function: Assesses the impact of deviations on the estimator.
- Breakdown point: Minimum fraction of contaminated data that compromises the estimator, with higher values indicating better robustness.

Sensitivity of Gaussian ML Estimators

- Gaussian-based estimators lack robustness and are highly sensitive to distribution tails.
- A single outlier can hinder the sample mean or covariance (breakdown point of $1/T$).

Robustness of Median and Heavy-Tailed ML Estimators

- The median offers greater robustness with a breakdown point of approximately 0.5.
- Heavy-tailed ML estimators effectively handle deviations, enhancing robustness.

Robust Estimators: M-Estimators*

Introduction to M-Estimators

- The term dates back to the 1960s, introduced by Huber (1964).
- They are a generalization of maximum likelihood estimators.
- They are defined by fixed-point equations for robust estimation of location and scatter.

Fixed-Point Equations for M-Estimators of $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$:

$$\frac{1}{T}\sum_{t=1}^{T} u_1\left(\sqrt{(\boldsymbol{x}_t-\boldsymbol{\mu})^\top\boldsymbol{\Sigma}^{-1}(\boldsymbol{x}_t-\boldsymbol{\mu})}\right)(\boldsymbol{x}_t-\boldsymbol{\mu}) = \mathbf{0}$$

$$\frac{1}{T}\sum_{t=1}^{T} u_2\left((\boldsymbol{x}_t-\boldsymbol{\mu})^\top\boldsymbol{\Sigma}^{-1}(\boldsymbol{x}_t-\boldsymbol{\mu})\right)(\boldsymbol{x}_t-\boldsymbol{\mu})(\boldsymbol{x}_t-\boldsymbol{\mu})^\top = \boldsymbol{\Sigma}$$

- The weight functions $u_1(\cdot)$ and $u_2(\cdot)$ must satisfy certain conditions.

Properties and Robustness of M-Estimators

- They are weighted sample mean and covariance matrix.
- They have a bounded influence function for robustness.
- The breakdown point is relatively low, despite robustness.

Robust Estimators: M-Estimators*

Other Robust Estimators (with higher breakdown points)

- Minimum volume ellipsoid (MVE).
- Minimum covariance determinant (MCD).

Gaussian ML Estimators as M-Estimators

- Trivial M-estimators with weight functions:

$$u_1(s) = u_2(s) = 1.$$

Relation to Heavy-Tailed ML Estimators

- M-estimators relate to heavy-tailed ML estimators with the choice:

$$u_1(s) = u_2(s^2) = \frac{\nu + N}{\nu + s^2}.$$

Robust Estimators: Tyler's Estimator

Tyler's Estimator for Scatter Matrix

- It was introduced by Tyler (1987) for heavy-tailed distributions.
- It is the most robust version of an M-estimator.

Elliptical Distribution Assumption

- It assumes a zero-mean elliptical distribution for $\boldsymbol{x}$.
- If the mean is not zero, it must be estimated and subtracted from the observations.

Normalization and ML Estimation

- Observations are normalized as
$$\boldsymbol{s}_t = \frac{\boldsymbol{x}_t}{\|\boldsymbol{x}_t\|_2}$$
- ML estimation is based on the normalized points, with the pdf (angular distribution) given by
$$f(\boldsymbol{s}) \propto \frac{1}{\sqrt{|\boldsymbol{\Sigma}|}} \left(\boldsymbol{s}^{\mathsf{T}} \boldsymbol{\Sigma}^{-1} \boldsymbol{s}\right)^{-N/2}$$

Robust Estimators: Tyler's Estimator

MLE Formulation

- Given T observations, the MLE is formulated as

$$\underset{\boldsymbol{\Sigma}}{\text{minimize}} \quad \log\det(\boldsymbol{\Sigma}) + \frac{N}{T}\sum_{t=1}^{T}\log\left(\boldsymbol{x}_t^\top\boldsymbol{\Sigma}^{-1}\boldsymbol{x}_t\right)$$

- This leads to the fixed-point equation

$$\boldsymbol{\Sigma} = \frac{1}{T}\sum_{t=1}^{T} w_t(\boldsymbol{\Sigma}) \times \boldsymbol{x}_t\boldsymbol{x}_t^\top,$$

  with weights given by

$$w_t(\boldsymbol{\Sigma}) = \frac{N}{\boldsymbol{x}_t^\top\boldsymbol{\Sigma}^{-1}\boldsymbol{x}_t}.$$

Robustness and Weights

- The weights enhance robustness by down-weighting outliers.
- A solution exists if $T > N$.

Numerical Experiments

Comparison of Estimators for Mean and Covariance Matrix

- Gaussian MLE
- Tyler's estimator for covariance matrix (paired with spatial median for location)
- Heavy-tailed MLE

Observations

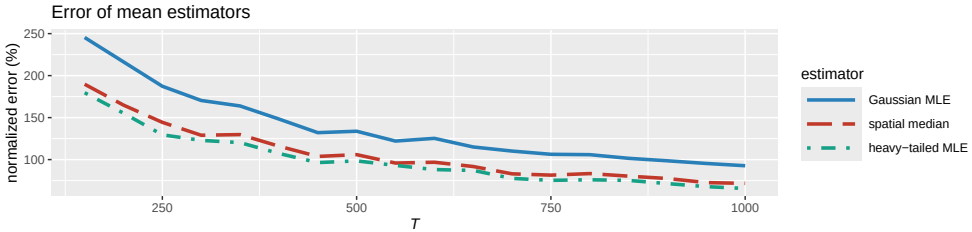
- Gaussian MLE and heavy-tailed MLE perform similarly for Gaussian tails.
- As tails become heavier (smaller ν), heavy-tailed MLE significantly outperforms Gaussian MLE.
- Tyler's estimator (with spatial median for the mean) is also not bad.

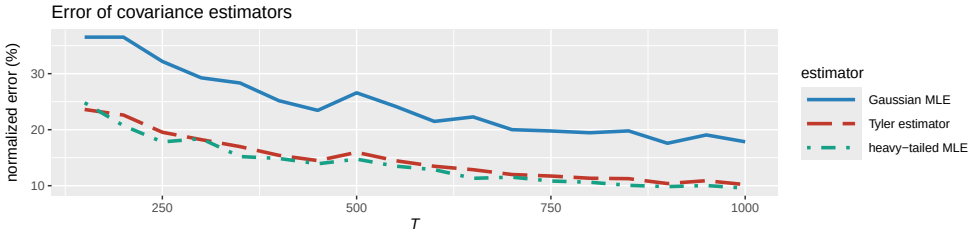
Conclusion

- Historical data-based mean vector $\boldsymbol{\mu}$ estimation errors can be substantial.
- Financial data's heavy-tailed nature necessitates robust heavy-tailed ML estimators.
- The computational cost of robust estimators is comparable to traditional sample estimators, with convergence in $3 \sim 5$ iterations.
- Practitioners often use factors from data providers for $\boldsymbol{\mu}$ estimation or opt for portfolio designs that bypass $\boldsymbol{\mu}$ estimation, such as GMVP or RPP.

Numerical Experiments

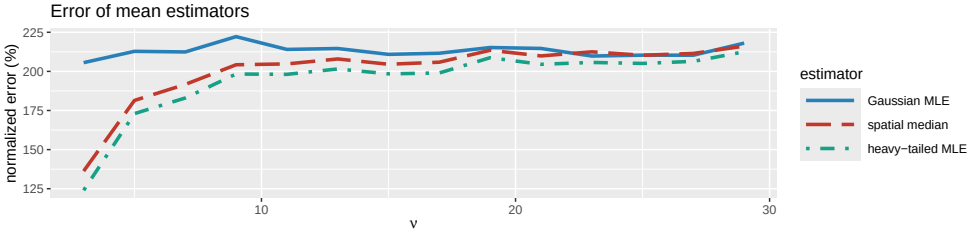
Estimation error of different ML estimators versus number of observations (for t-distributed heavy-tailed data with $\nu = 4$ and $N = 100$):

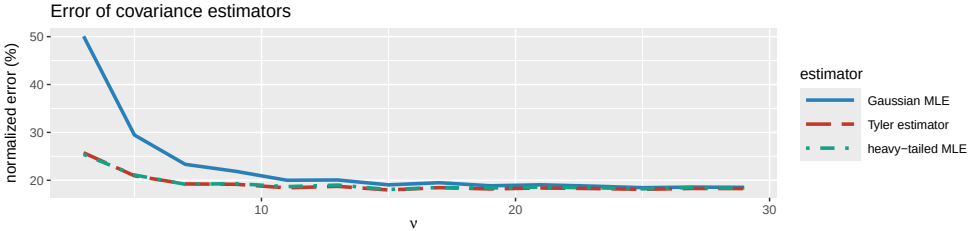




Numerical Experiments

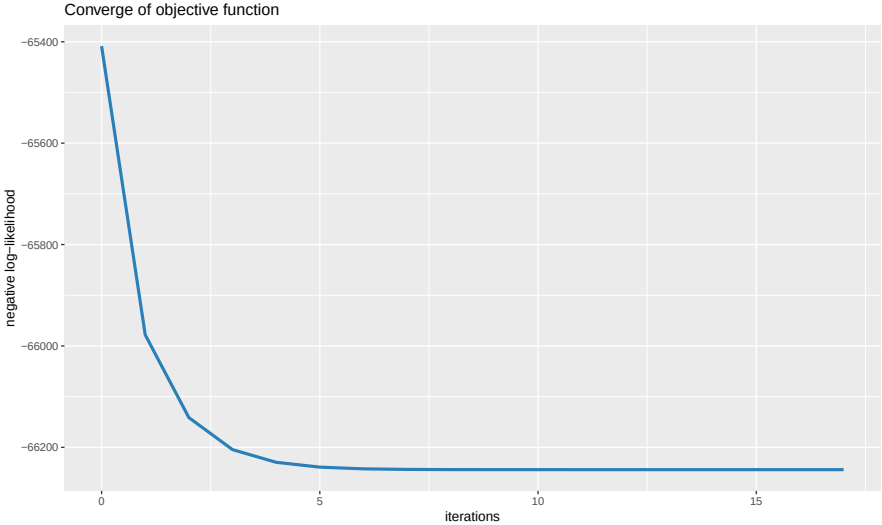
Estimation error of different ML estimators versus degrees of freedom in a t distribution (with $T = 200$ and $N = 100$):





Numerical Experiments

Convergence of robust heavy-tailed ML estimator:



Outline

1. I.I.D. Model
2. Sample Estimators
3. Location Estimators
4. Gaussian ML Estimators
5. Heavy-Tailed ML Estimators
6. Prior Information
   - Shrinkage
   - Factor Models
   - Black-Litterman Model
7. Summary

Prior Information

Challenges With Historical Data

- A limited number of observations (T) can lead to high estimation errors.
- Practical settings often lack sufficient data for accurate parameter estimation.

Improving Estimators With Prior Information

- Researchers and practitioners have developed methods to integrate prior knowledge to enhance estimators.

Three Popular Methods to Incorporate Prior Information

- **Shrinkage** integrates prior knowledge through parameter targets and aims to improve estimation by pulling estimates towards a target.
- **Factor modeling** utilizes structural information about the data, which helps in reducing dimensionality and improving parameter estimation.
- The **Black-Litterman** approach merges historical data with subjective views, balancing empirical data with investor-specific insights.

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Shrinkage

Introduction to Shrinkage

- This is a technique to reduce estimation error by introducing bias.
- It originated with Stein in 1955 and has been popular in finance for covariance matrix shrinkage since the early 2000s.

Bias-Variance Trade-off

- The mean squared error (MSE) of an estimator is the sum of its variance and squared bias.
- Small sample sizes lead to high variance, while larger samples may see bias dominate the error.

Stein's Seminal Contribution (Stein 1955)

- Demonstrated the benefit of introducing bias for overall error reduction.
- Shrinkage involves moving an estimator towards a target value to minimize error.

Shrinkage Estimator Formula

$$\hat{\boldsymbol{\theta}}^{\mathrm{sh}} = (1-\rho)\,\hat{\boldsymbol{\theta}} + \rho\,\boldsymbol{\theta}^{\mathrm{tgt}}$$

where ρ is the shrinkage factor, $\hat{\boldsymbol{\theta}}$ denotes the original estimator, and $\boldsymbol{\theta}^{\mathrm{tgt}}$ is the target.

Shrinkage

Implementation Considerations

- The **choice of target** ($\boldsymbol{\theta}^{\text{tgt}}$) represents prior information or market views.
- The **choice of shrinkage factor** (ρ) is critical for balancing the weight of the target in the estimator.

Choosing the Shrinkage Factor

- An **empirical choice** is based on cross-validation.
- An **analytical choice** utilizes advanced mathematical techniques (e.g., random matrix theory).

Application in Finance

- The parameter $\boldsymbol{\theta}$ could be the mean vector $\boldsymbol{\mu}$ or the covariance matrix $\boldsymbol{\Sigma}$.
- Estimators like the sample mean or covariance matrix can be adjusted using shrinkage for better accuracy.

Shrinking the Mean Vector

Sample Mean and Its Properties

- The sample mean $\hat{\boldsymbol{\mu}}$ is an unbiased estimator of the mean vector $\boldsymbol{\mu}$.
- Its distribution is characterized by $\hat{\boldsymbol{\mu}} \sim \mathcal{N}\left(\boldsymbol{\mu}, \frac{1}{T}\boldsymbol{\Sigma}\right)$.
- The mean squared error (MSE) is $\mathbb{E}\left[\left\|\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}\right\|^2\right] = \frac{1}{T}\mathrm{Tr}(\boldsymbol{\Sigma})$.

Stein's Insight on Bias and MSE

- Stein's 1955 paper (Stein 1955) showed that allowing bias can reduce the overall MSE.
- Shrinkage introduces bias towards a target to achieve a lower MSE.

James-Stein Estimator

$$\hat{\boldsymbol{\mu}}^{\mathrm{JS}} = (1-\rho)\,\hat{\boldsymbol{\mu}} + \rho\,\boldsymbol{\mu}^{\mathrm{tgt}}$$

- It improves the MSE over the sample mean for any target $\boldsymbol{\mu}^{\mathrm{tgt}}$ with a properly chosen ρ:

$$\rho = \frac{(N+2)}{(N+2) + T \times \left(\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}^{\mathrm{tgt}}\right)^{\mathsf{T}} \boldsymbol{\Sigma}^{-1} \left(\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}^{\mathrm{tgt}}\right)}$$

Shrinking the Mean Vector

Adaptability of ρ

- $\rho \to 0$ as T increases, favoring the original sample mean.
- $\rho \to 0$ if the target significantly differs from the sample mean, acting as a safety mechanism.

Choosing the Target $\boldsymbol{\mu}^{\mathbf{tgt}}$

- The choice of target is flexible, but MSE improvement depends on the target's informativeness.
- Common choices include:
  - Zero: $\boldsymbol{\mu}^{\text{tgt}} = \mathbf{0}$.
  - Grand mean: $\boldsymbol{\mu}^{\text{tgt}} = \frac{\mathbf{1}^\top \hat{\boldsymbol{\mu}}}{N} \times \mathbf{1}$.
  - Volatility-weighted grand mean: $\boldsymbol{\mu}^{\text{tgt}} = \frac{\mathbf{1}^\top \hat{\boldsymbol{\Sigma}}^{-1} \hat{\boldsymbol{\mu}}}{\mathbf{1}^\top \hat{\boldsymbol{\Sigma}}^{-1} \mathbf{1}} \times \mathbf{1}$.

Shrinking the Covariance Matrix

Shrinkage in Covariance Matrix Estimation

- Introduces bias to reduce estimation error.
- The shrinkage estimator formula is:

$$\hat{\boldsymbol{\Sigma}}^{\text{sh}} = (1-\rho)\,\hat{\boldsymbol{\Sigma}} + \rho\,\boldsymbol{\Sigma}^{\text{tgt}},$$

  where $\boldsymbol{\Sigma}^{\text{tgt}}$ is the target and ρ is the shrinkage factor.

Historical Context

- The concept was used in the 1980s in wireless communications as "diagonal loading".
- It gained popularity in finance in the early 2000s through the work of Ledoit and Wolf (Ledoit and Wolf 2003, 2004).

Common Targets for Covariance Matrix

- Scaled identity matrix: $\boldsymbol{\Sigma}^{\text{tgt}} = \frac{1}{N}\text{Tr}(\hat{\boldsymbol{\Sigma}}) \times \boldsymbol{I}$.
- Diagonal matrix: $\boldsymbol{\Sigma}^{\text{tgt}} = \text{Diag}(\hat{\boldsymbol{\Sigma}})$.
- Equal-correlation matrix, where the off-diagonal elements are equal to the average cross-correlation.

Shrinking the Covariance Matrix

Determining the Shrinkage Factor ρ

- Can be empirical (via cross-validation) or analytical: random matrix theory (RMT).
- Ledoit and Wolf popularized the RMT approach to minimize $\mathbb{E}\left[\left\|\hat{\boldsymbol{\Sigma}}^{\mathrm{sh}} - \boldsymbol{\Sigma}\right\|_{\mathrm{F}}^2\right]$.
- The asymptotic RMT formula for ρ is (Ledoit and Wolf 2003, 2004):

$$\rho = \min\left(1, \frac{\frac{1}{T}\sum_{t=1}^{T}\left\|\hat{\boldsymbol{\Sigma}} - \boldsymbol{x}_t\boldsymbol{x}_t^{\mathrm{T}}\right\|_{\mathrm{F}}^2}{\left\|\hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}^{\mathrm{tgt}}\right\|_{\mathrm{F}}^2}\right).$$

Extension to Heavy-Tailed Distributions

- Shrinkage factor ρ can be adapted for heavy-tailed distributions, enhancing robustness.

Alternative Error Measures

- Error in terms of the inverse covariance matrix: $\mathbb{E}\left[\left\|\left(\hat{\boldsymbol{\Sigma}}^{\mathrm{sh}}\right)^{-1} - \boldsymbol{\Sigma}^{-1}\right\|_{\mathrm{F}}^2\right]$.
- Choose ρ to maximize directly the achieved Sharpe ratio.

Nonlinear Shrinkage

- Extends the idea to the eigenvalues of the covariance matrix.
- Requires increased mathematical sophistication.

Numerical Experiments

Shrinkage Estimators and Observation Size

- Estimation error analysis for shrinkage estimators with synthetic Gaussian data.
- Clear improvement in mean vector estimation; modest improvement in covariance matrix estimation.
- Shrinkage benefits decrease as the number of observations (T) increases.

Shrinkage to Zero

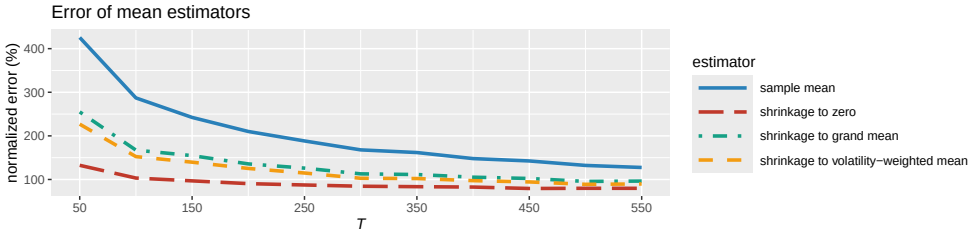
- Shrinkage to zero return vector yields the best results.
- Aligns with the efficient-market hypothesis, suggesting current prices reflect all available information.

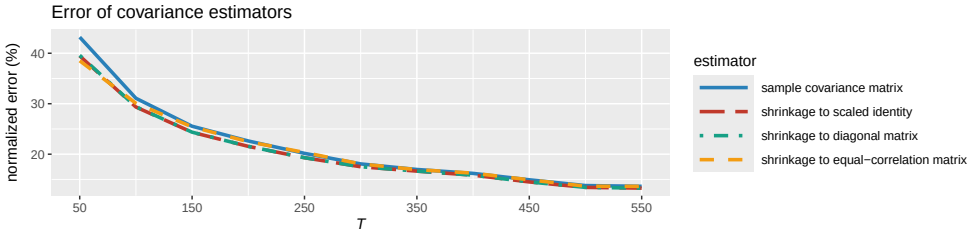
Consideration of MSE in Portfolio Optimization

- Numerical results are based on the MSE of estimators.
- MSE may not be the optimal error measure in portfolio optimization contexts.
- The importance of using more appropriate error measures is highlighted.

Numerical Experiments

Estimation error of different shrinkage estimators versus number of observations (for Gaussian data with $N = 100$):





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Factor Models

Factor Modeling in Finance

- Incorporates prior information into asset return models.
- Found in numerous finance textbooks.

Single-Factor Model

- The simplest form of factor modeling.
- The equation is:

$$\boldsymbol{x}_t = \boldsymbol{\alpha} + \boldsymbol{\beta} f_t^{\text{mkt}} + \boldsymbol{\epsilon}_t$$

- $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ represent asset-specific alpha and beta.
- f_t^{mkt} is the market factor and $\boldsymbol{\epsilon}_t$ is the zero-mean residual.

Connection to CAPM

- The single-factor model is related to the Capital Asset Pricing Model (CAPM).
- CAPM assumes zero alpha and relates expected excess returns to market beta.

Factor Models

Multi-Factor Modeling

- A generalization of the single-factor model.
- The equation is:

$$\boldsymbol{x}_t = \boldsymbol{\alpha} + \boldsymbol{B}\boldsymbol{f}_t + \boldsymbol{\epsilon}_t$$

- $\boldsymbol{f}_t$ contains K factors and $\boldsymbol{B}$ has factor loadings.
- Dynamic factor models include time-dependency in the factors.

Idiosyncratic Component

- The residual term $\boldsymbol{\epsilon}_t$ is assumed to have a diagonal covariance matrix $\boldsymbol{\Psi}$.
- It captures asset-specific noise not explained by the factors.

Factor Models

Mean and Covariance Matrix:

$$\boldsymbol{\mu} = \boldsymbol{\alpha} + \boldsymbol{B}\boldsymbol{\mu}_f$$

$$\boldsymbol{\Sigma} = \boldsymbol{B}\boldsymbol{\Sigma}_f\boldsymbol{B}^\mathsf{T} + \boldsymbol{\Psi}$$

- Decomposes the covariance into low-rank and full-rank diagonal components.
- For a single-factor model, the covariance matrix is: $\boldsymbol{\Sigma} = \sigma_f^2\boldsymbol{\beta}\boldsymbol{\beta}^\mathsf{T} + \boldsymbol{\Psi}$.

Parameter Reduction

- Factor models reduce the number of parameters that need to be estimated.
- For example, for $N = 500$ assets and $K = 3$ factors, the number of parameters is reduced from 125,750 to 1,503.

Types:

- **Macroeconomic factor models** use observable economic factors and have unknown loadings.
- **Fundamental factor models** use loadings from asset characteristics and have unknown factors.
- **Statistical factor models** have both unknown factors and loadings.

Macroeconomic Factor Models

Macroeconomic Factor Models Overview

- Allows for the integration of economic indicators into the analysis of asset returns.
- Utilizes observable economic time series as factors (e.g., market index, GDP growth rate, interest rates, inflation rates).
- Factors are often proprietary, derived from complex analyses of various data sources.
- Investment funds may pay high premiums for access to these factors, whereas small investors might use publicly available data.

Parameter Estimation

- With known factors, model parameters ($\boldsymbol{\alpha}$ and $\boldsymbol{B}$) can be estimated through least squares regression:

$$\underset{\boldsymbol{\alpha},\boldsymbol{B}}{\text{minimize}} \sum_{t=1}^{T} \|\boldsymbol{x}_t - (\boldsymbol{\alpha} + \boldsymbol{B}\boldsymbol{f}_t)\|_2^2 ,$$

- The mean vector $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$ are derived from the estimated parameters as per the factor model equation.

Fundamental Factor Models

Fundamental Factor Models Overview

- Use observable asset characteristics, known as fundamentals, to define factors.
- Common fundamentals include industry classification, market capitalization, and style classification (value, growth).

Industry Approaches

- Fama-French approach:
  - Form portfolios based on asset characteristics to derive factors $\boldsymbol{f}_t$.
  - Loadings $\boldsymbol{B}$ are estimated similarly to macroeconomic models.
  - The original model had $K = 3$ factors: firm size, book-to-market values, and excess market return (Fama and French 1992).
  - It was extended to $K = 5$ factors, including profitability and investment patterns (Fama and French 2015).
- Barra risk factor analysis approach:
  - Loadings $\boldsymbol{B}$ are constructed from asset characteristics.
  - Factors $\boldsymbol{f}_t$ are estimated via regression, opposite of macroeconomic models.
  - This approach was developed by Barra Inc. in 1975.

Statistical Factor Models

Statistical Factor Models Overview

- Both the factors $\boldsymbol{f}_t$ and the loading matrix $\boldsymbol{B}$ are unknown.
- They introduce structure to the covariance matrix $\boldsymbol{\Sigma}$ as a low-rank plus diagonal matrix.

Covariance Matrix Structure

- $\boldsymbol{\Sigma}$ is decomposed into $\boldsymbol{B}\boldsymbol{\Sigma}_f\boldsymbol{B}^\mathsf{T}$ (low-rank) and $\boldsymbol{\Psi}$ (diagonal).
- The factors are assumed to be zero-mean and normalized for simplification.

Heuristic Formulation

Approximate the sample covariance matrix $\hat{\boldsymbol{\Sigma}}$ with the desired structure:

$$\underset{\boldsymbol{B},\boldsymbol{\psi}}{\text{minimize}} \quad \|\hat{\boldsymbol{\Sigma}} - \left(\boldsymbol{B}\boldsymbol{B}^\mathsf{T} + \mathrm{Diag}(\boldsymbol{\psi})\right)\|_\mathsf{F}^2.$$

Statistical Factor Models

ML Estimation Under the Gaussian Assumption

- The ML estimation is formulated by imposing the covariance structure:

$$\begin{array}{ll} \underset{\boldsymbol{\alpha},\boldsymbol{\Sigma},\boldsymbol{B},\boldsymbol{\psi}}{\text{minimize}} & \log\det(\boldsymbol{\Sigma}) + \dfrac{1}{T}\displaystyle\sum_{t=1}^{T}(\boldsymbol{x}_t - \boldsymbol{\alpha})^{\mathsf{T}}\boldsymbol{\Sigma}^{-1}(\boldsymbol{x}_t - \boldsymbol{\alpha}) \\ \text{subject to} & \boldsymbol{\Sigma} = \boldsymbol{B}\boldsymbol{B}^{\mathsf{T}} + \mathrm{Diag}(\boldsymbol{\psi}). \end{array}$$

- The nonconvex nature of the problem makes it challenging.
- Iterative algorithms have been developed to address the nonconvex optimization challenges.

Extensions

- Heavy-tailed distributions can be accommodated.
- The structure of financial data, such as nonnegative asset correlation, can be integrated into model formulations.

Principal Component Analysis (PCA)*

PCA in a Nutshell

- It is a statistical technique for handling high-dimensional datasets by focusing on the most significant variance components.
- It facilitates more efficient data analysis and interpretation.

PCA for Dimension Reduction

- PCA identifies the directions of maximum variance in high-dimensional data.
- It simplifies data analysis by reducing dimensions to a lower-dimensional subspace.

PCA Methodology

- It maximizes the variance along a direction $\boldsymbol{u}$: $\mathrm{Var}(\boldsymbol{u}^\top \boldsymbol{x}) = \boldsymbol{u}^\top \boldsymbol{\Sigma} \boldsymbol{u}$.
- The solution is found through eigenvalue decomposition: $\boldsymbol{\Sigma} \approx \boldsymbol{U}^{(K)} \boldsymbol{D}^{(K)} \boldsymbol{U}^{(K)\top}$.
- $\boldsymbol{U}^{(K)}$ contains the first K eigenvectors, and $\boldsymbol{D}^{(K)}$ has the largest K eigenvalues.

Principal Component Analysis (PCA)*

PCA in Statistical Factor Models

- It approximates the solution to the statistical factor model by performing PCA on the sample covariance matrix.
- As a heuristic, it keeps K principal components and uses a scaled identity matrix for the diagonal component $\boldsymbol{\Psi}$.
- The approximate solution to the factor model is:

$$\boldsymbol{B} = \boldsymbol{U}^{(K)}\text{Diag}\left(\sqrt{\lambda_1 - \kappa}, \ldots, \sqrt{\lambda_K - \kappa}\right)$$

$$\boldsymbol{\Psi} = \kappa \boldsymbol{I}$$

  where κ is the average of the $N - K$ smallest eigenvalues.

PCA Estimator for the Covariance Matrix

- The PCA estimator for the covariance matrix is:

$$\hat{\boldsymbol{\Sigma}} = \boldsymbol{U}\text{Diag}\left(\lambda_1, \ldots, \lambda_K, \kappa, \ldots, \kappa\right)\boldsymbol{U}^{\mathsf{T}}.$$

- This achieves noise averaging of the smallest eigenvalues, similar to the concept of shrinkage.

Numerical Experiments

Evaluation of Covariance Matrix Estimation Under a Factor Model

- Estimation accuracy depends on how well the data follows the factor model structure.
- Incorrect assumptions about the factor model structure can lead to poorer results than using the sample covariance matrix.

Importance of Model Choice

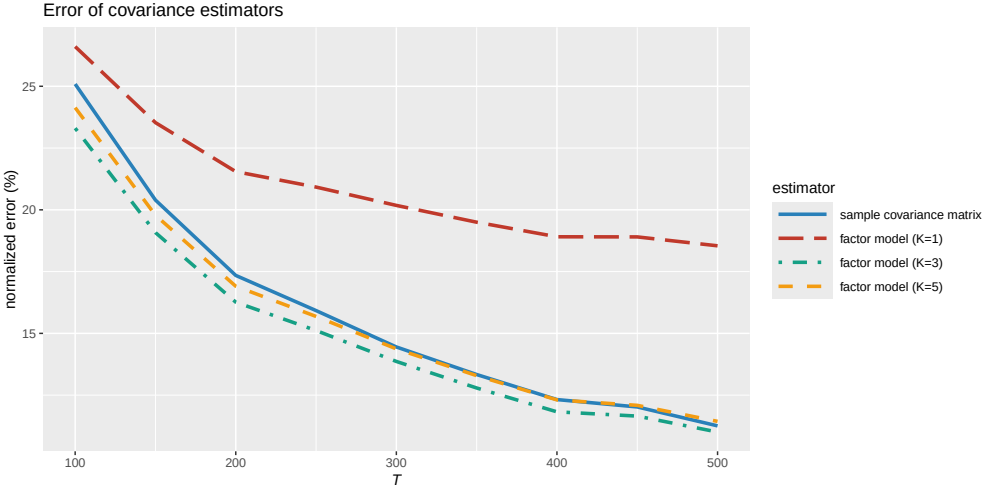
- The decision to use a factor model for covariance matrix estimation must be made cautiously.
- Factor modeling strategies and their implications on trading are elaborated in (Fabozzi, Focardi, and Kolm 2010).

Observations From Synthetic Data Analysis

- An estimation error comparison between factor model-based and sample covariance matrices is shown for synthetic Gaussian data.
- The data complies with a factor model structure having $K = 3$ principal components.
- Accurate estimation with the correct K improves results over the sample covariance matrix.
- Incorrect K values, such as $K = 1$, can significantly worsen estimation accuracy.

Numerical Experiments

Estimation error of covariance matrix under factor modeling versus number of observations (with $N = 100$):



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Black-Litterman Model

Black-Litterman Model Basics

- It combines historical data with an investor's prior information.
- It is a standard in finance, detailed in textbooks like (Fabozzi, Focardi, and Kolm 2010) and (Meucci 2005).

Components of the Black-Litterman Model

- **Market equilibrium** is an estimate of $\boldsymbol{\mu}$ from the market, denoted as $\boldsymbol{\pi} = \hat{\boldsymbol{\mu}}$:

$$\boldsymbol{\pi} = \boldsymbol{\mu} + \boldsymbol{\epsilon}$$

  with $\boldsymbol{\epsilon}$ being zero-mean and having a covariance of $\tau\boldsymbol{\Sigma}$, where τ is often set to $1/T$.
- **Investor's views** consist of K views on asset returns, expressed as:

$$\boldsymbol{v} = \boldsymbol{P}\boldsymbol{\mu} + \boldsymbol{e}$$

  where $\boldsymbol{v}$ and $\boldsymbol{P}$ represent the views and their relation to asset returns, and the error term $\boldsymbol{e}$ is zero-mean with covariance $\boldsymbol{\Omega}$.

Views

Quantitative and Qualitative Views

- Quantitative views specify expected returns and their uncertainties.
- Qualitative views provide directional expectations (e.g., bullish, bearish) without specific return figures.

Example of Quantitative Investor's Views Two independent views on $N = 5$ stocks:

- View 1: stock 1 expected to return 1.5% with a standard deviation of 1%.
- View 2: stock 3 expected to outperform stock 2 by 4% with a 1% standard deviation.

Mathematical Representation

- The views are expressed as:

$$\begin{bmatrix} 1.5\% \\ 4\% \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & -1 & 1 & 0 & 0 \end{bmatrix} \boldsymbol{\mu} + \boldsymbol{e}$$

- The covariance of the error $\boldsymbol{e}$ is:

$$\boldsymbol{\Omega} = \begin{bmatrix} 1\%^2 & 0 \\ 0 & 1\%^2 \end{bmatrix}.$$

Merging the Market Equilibrium with the Views

Combining Market Equilibrium and Investor's Views

- Various mathematical formulations (least squares, maximum likelihood, Bayesian) yield similar solutions for integrating market equilibrium with an investor's views.

Weighted Least Squares Formulation

- A compact representation is:

$$\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\mu} + \boldsymbol{n},$$

  where $\boldsymbol{y} = \begin{bmatrix} \boldsymbol{\pi} \\ \boldsymbol{v} \end{bmatrix}$, $\boldsymbol{X} = \begin{bmatrix} \boldsymbol{I} \\ \boldsymbol{P} \end{bmatrix}$, and the noise covariance is $\boldsymbol{V} = \begin{bmatrix} \tau\boldsymbol{\Sigma} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{\Omega} \end{bmatrix}$.
- The problem is formulated as:

$$\underset{\boldsymbol{\mu}}{\text{minimize}} \quad (\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\mu})^{\mathsf{T}} \boldsymbol{V}^{-1} (\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\mu}),$$

- The solution is:

$$\begin{aligned} \boldsymbol{\mu}_{\mathrm{BL}} &= \left(\boldsymbol{X}^{\mathsf{T}}\boldsymbol{V}^{-1}\boldsymbol{X}\right)^{-1} \boldsymbol{V}^{-1}\boldsymbol{y} \\ &= \left((\tau\boldsymbol{\Sigma})^{-1} + \boldsymbol{P}^{\mathsf{T}}\boldsymbol{\Omega}^{-1}\boldsymbol{P}\right)^{-1} \left((\tau\boldsymbol{\Sigma})^{-1}\boldsymbol{\pi} + \boldsymbol{P}^{\mathsf{T}}\boldsymbol{\Omega}^{-1}\boldsymbol{v}\right) \end{aligned}$$

Merging the Market Equilibrium with the Views*

Original Bayesian Formulation of the Black-Litterman Model

- Returns $\boldsymbol{x}$ are assumed to follow a normal distribution: $\boldsymbol{x} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$.
- The mean $\boldsymbol{\mu}$ is also modeled as random with a Gaussian distribution, $\boldsymbol{\mu} \sim \mathcal{N}(\boldsymbol{\pi}, \tau\boldsymbol{\Sigma})$, where $\boldsymbol{\pi}$ is the best guess for $\boldsymbol{\mu}$ and $\tau\boldsymbol{\Sigma}$ represents the uncertainty.
- Views are modeled with a Gaussian distribution: $\boldsymbol{P}\boldsymbol{\mu} \sim \mathcal{N}(\boldsymbol{v}, \boldsymbol{\Omega})$.
- The posterior distribution of $\boldsymbol{\mu}$ given the views is $\boldsymbol{\mu} \mid \boldsymbol{v}, \boldsymbol{\Omega} \sim \mathcal{N}\left(\boldsymbol{\mu}_{\mathrm{BL}}, \boldsymbol{\Sigma}_{\mathrm{BL}}\right)$.
- The posterior mean $\boldsymbol{\mu}_{\mathrm{BL}}$ matches the weighted least squares solution.
- The posterior covariance $\boldsymbol{\Sigma}_{\mathrm{BL}}$ includes the original covariance $\boldsymbol{\Sigma}$ and the covariance of the posterior mean.
- The mean estimator is:

$$\boldsymbol{\mu}_{\mathrm{BL}} = \boldsymbol{\pi} + \tau\boldsymbol{\Sigma}\boldsymbol{P}^{\mathsf{T}}\left(\tau\boldsymbol{P}\boldsymbol{\Sigma}\boldsymbol{P}^{\mathsf{T}} + \boldsymbol{\Omega}\right)^{-1}(\boldsymbol{v} - \boldsymbol{P}\boldsymbol{\pi})$$

- The covariance matrix estimator is:

$$\boldsymbol{\Sigma}_{\mathrm{BL}} = (1+\tau)\boldsymbol{\Sigma} - \tau^2\boldsymbol{\Sigma}\boldsymbol{P}^{\mathsf{T}}\left(\tau\boldsymbol{P}\boldsymbol{\Sigma}\boldsymbol{P}^{\mathsf{T}} + \boldsymbol{\Omega}\right)^{-1}\boldsymbol{P}\boldsymbol{\Sigma}.$$

Merging the Market Equilibrium with the Views*

Alternative Bayesian Formulation

- A variation introduced by (Meucci 2005) models views on random returns as $\boldsymbol{v} = \boldsymbol{P}\boldsymbol{x} + \boldsymbol{e}$, which differs from the original formulation where views are on $\boldsymbol{\mu}$.
- This approach leads to a posterior distribution of returns with results akin to the original Black-Litterman model.

Impact of Parameter τ on the Black-Litterman Estimator

- For $\tau = 0$, the market equilibrium is considered completely accurate, leading to $\boldsymbol{\mu}_{\mathrm{BL}} = \boldsymbol{\pi}$.
- As $\tau \to \infty$, the market equilibrium is disregarded, and the investor's views solely influence the outcome, resulting in $\boldsymbol{\mu}_{\mathrm{BL}} = \left(\boldsymbol{P}^{\mathsf{T}}\boldsymbol{\Omega}^{-1}\boldsymbol{P}\right)^{-1}\left(\boldsymbol{P}^{\mathsf{T}}\boldsymbol{\Omega}^{-1}\boldsymbol{v}\right)$.
- For $0 < \tau < \infty$, $\boldsymbol{\mu}_{\mathrm{BL}}$ represents a blend of the market equilibrium and investor's views, embodying the principle of shrinkage.

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Summary

Countless models for financial data exist in the literature. The i.i.d. model, while rough, is functional and widely used. Key points of the i.i.d. model for financial data include:

- **Sample estimators perform poorly** since Gaussian assumptions don't hold in practice.
- **Robust estimators are necessary** to handle outliers, like spatial median and Tyler estimator.
- **Heavy-tailed estimators suit financial data well** as they are naturally robust. Simple iterative algorithms can compute them.
- **Estimating mean vector from historical data is extremely noisy**. Practitioners use premium data provider factors for regression instead.
- **Prior information should be used when available** via shrinkage, factor modeling, or Black-Litterman information fusion.

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